Device Type Classification on ISP Network using Time Series Analysis

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MACHINE LEARNING | NETWORK TRAFFIC | ISP SECURITY



Device Diversity at ISP Scale

Why It Matters

ISP networks operate at up to **400 Gbps**, connecting a vast array of devices. Accurately identifying these types is critical for QoS, security, and optimization.

Key challenges include:

- Enabling better QoS & security policy enforcement
- (D)DoS protection & bandwidth optimization
- Handling massive traffic volumes beyond enterprise scale

End-Devices

Laptops, smartphones, workstations, and multiple workstations inside NAT IP addresses.

Net-Devices

Routers and switches providing critical backbone connectivity.

Servers

Hosting and cloud infrastructure requiring high bandwidth and constant uptime.

This is the first comprehensive study at the ISP level, where traffic diversity is substantially higher.



Key Contributions

1. **Novel Method**
   Time-series-based device type recognition at ISP scale, achieving **over 90% classification accuracy**

2. **Temporal Stability**
   Extensive analysis across **12 test weeks** to evaluate data drift effects for real-world ML deployment

3. **Annotated Dataset**
   Device-type annotations for CESNET-TimeSeries-24, published via CESNET TS-Zoo and Zenodo

4. **Open Source**
   Complete implementation, datasets, and experimental pipelines released publicly on GitHub

Dataset: CESNET-TimeSeries-24

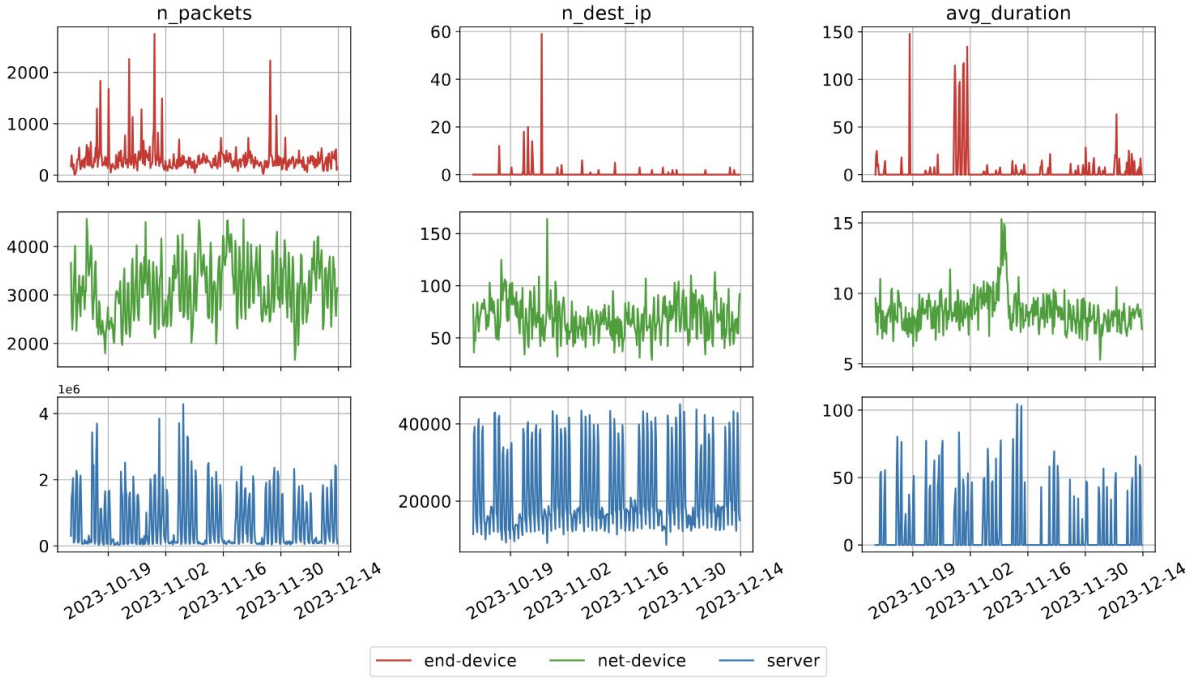
The dataset captures traffic from the CESNET network as multivariate time series. This study uses the **1-hour aggregation**.

Labels were assigned to **82,504 devices** via semi-automated annotation.

87.9% **End-Device**
72,523 devices

9.5% **Server**
7,875 devices

2.6% **Net-Device**
2,106 IPs



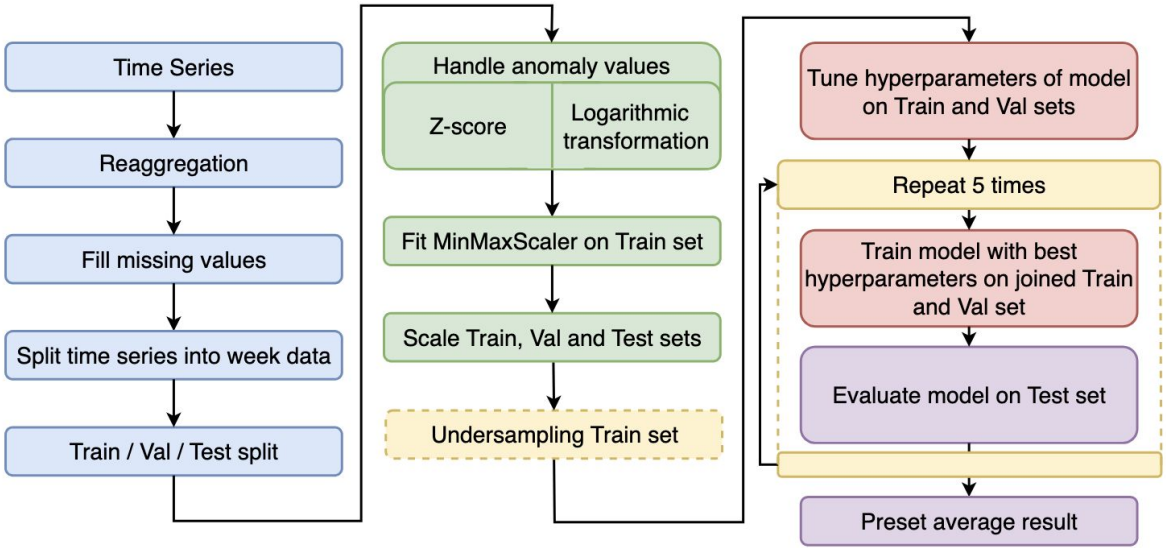
Methodology Pipeline

1 Reaggregation
1-hour → 4-hour intervals to reduce model load while preserving temporal patterns

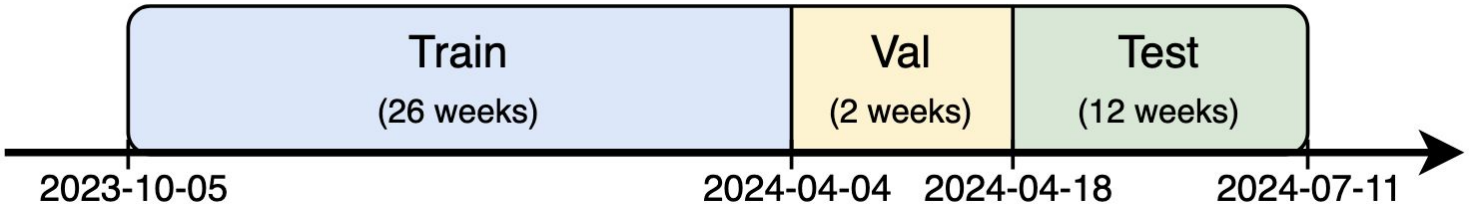
2 Fill & Split
Missing values filled with zeros; time series split into weekly windows to capture periodic behavior

3 Anomaly Handling
Log transformation or Z-score (>4 threshold, removes ~15% of training data); Min-Max scaling applied

4 Train/Val/Test Split
26 weeks train, 2 weeks val, 12 weeks test — temporal splitting to assess generalization and data drift



Temporal Data Split & Feature Subsets



TRAIN SET
2,401,854 samples
26 weeks

VALIDATION SET
184,758 samples
2 weeks

TEST SET
1,108,548 samples
12 weeks

Feature Subset	Features Used
FS 1	All features
FS 2	n_flows, n_packets, n_bytes
FS 3	n_bytes, n_dest_ports, n_dest_ip, dir_ratio_bytes, avg_ttl
FS 4	tcp_udp_ratio_bytes, dir_ratio_bytes, avg_duration, avg_ttl

Model Performance Results

Model	Accuracy	Macro F1	Recall	Precision
K-means (TS)	0.915	0.676	0.654	0.735
K-means (H)	0.917	0.685	0.655	0.754
MLP	0.923	0.740	0.715	0.778
RNN	0.926	0.754	0.730	0.789
LSTM	0.932	0.769	0.748	0.803
GRU (Best)	**0.933**	**0.771**	**0.743**	**0.813**

GRU Performance Lead

LSTM trails closely by only 0.004 F1 and 0.001 accuracy. Recurrent models significantly outperform baselines.

93.3%

Peak Accuracy (GRU)

77.1%

Max Macro F1 (GRU)

GRU Per-Class Analysis

Class	Precision	Recall	F1-score
End device	0.951	0.982	**0.966**
Server	0.774	0.549	0.643
Net device	0.711	0.699	0.705
Macro avg	0.812	0.743	0.771
Weighted avg	0.928	0.933	0.929

Table VI: GRU classification report averaged over 5 runs.

End-Device: Strong ✓

Dominant class (87.9%) yields a stable decision boundary — F1 of **0.966**

Server: Challenging ✗

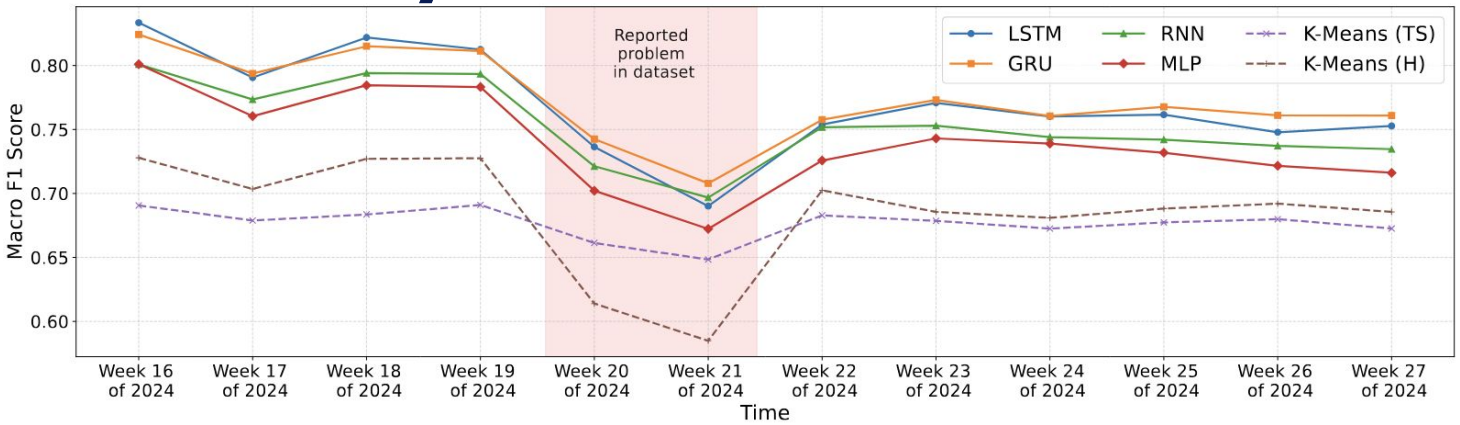
Low recall (0.549) — sporadic servers overlap with end-device traffic patterns

Net-Device: Balanced ✓

Consistent F1 of **0.705** despite limited representation in the dataset

Temporal Stability & Data Drift

All models exhibit a **general downward trend** in Macro F1-score across 12 test weeks, confirming incremental data drift.



Model	First Week F1	Last Week F1	Delta
K-means (TS)	0.690	0.672	**-0.018**
K-means (H)	0.727	0.685	-0.042
MLP	0.801	0.716	-0.085
LSTM	**0.833**	0.752	-0.081
GRU	0.824	**0.760**	-0.064

Probe Failure: Weeks 20–21

A sharp drop corresponds to a reported monitoring probe failure, causing incomplete data capture.

Operational Robustness

GRU maintains the highest absolute F1 throughout the test period despite drift.

Conclusions & Future Work

Feasibility Proven

Time-series ML classification is effective at ISP scale. GRU achieves **93.3% accuracy** and **77.1% Macro F1**, outperforming all baselines.

Drift Is Real

Incremental data drift degrades all models over time. Rapid drops correlate with infrastructure anomalies — robust preprocessing is critical.

Future Directions

Enhance drift detection, develop adaptive retraining strategies, and improve handling of missing data for greater operational reliability.

All code, annotated datasets (CESNET-DeviceType24 on Zenodo), and experimental pipelines are released as **open-source** to enable reproducibility and broader adoption in the network management research community.